



B.SC COMPUTER SCIENCE | GUIDE: DR. A.M. HEMA

Smart TASMACH – Consumer Regulation System

Aadhaar-Based Digital Alcohol Purchase Regulation Platform

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Problem Statement

No Identity Verification

- Purchases occur without Aadhaar or reliable ID checks.

Underage & Unlimited Buying

- Minors can buy alcohol and consumers face no limits.

No Monitoring or Analytics

- No purchase history, doctor restrictions, family monitoring, or real-time government analytics.

Lack of Transparency

- Manual reporting and absence of digital audit reduce accountability.



Existing System — Workflow



The current TASMAC retail workflow lacks digital identity checks, audit trails, health monitoring, purchase limits, and government analytics.

Proposed Solution Overview



Aadhaar Registration

Secure, verified consumer identities.



QR Code Identity

Quick in-shop verification.



Purchase Limits

Enforce consumption caps per consumer.



Doctor Restrictions

Medical professionals can restrict purchases.

Features & Benefits



Government Dashboard

Real-time analytics and reports for policy and enforcement.



Notifications & Digital Audit

Automated alerts and maintainable purchase history (PDF reports).



Caretaker Monitoring

Family members can supervise and enable teetotaler mode.

Project Objectives

Prevent Underage Purchase

Identity verification blocks minors.

Regulate Consumption

Purchase limits and teetotaler options promote responsibility.

Support Healthcare

Doctors can restrict patients as needed.

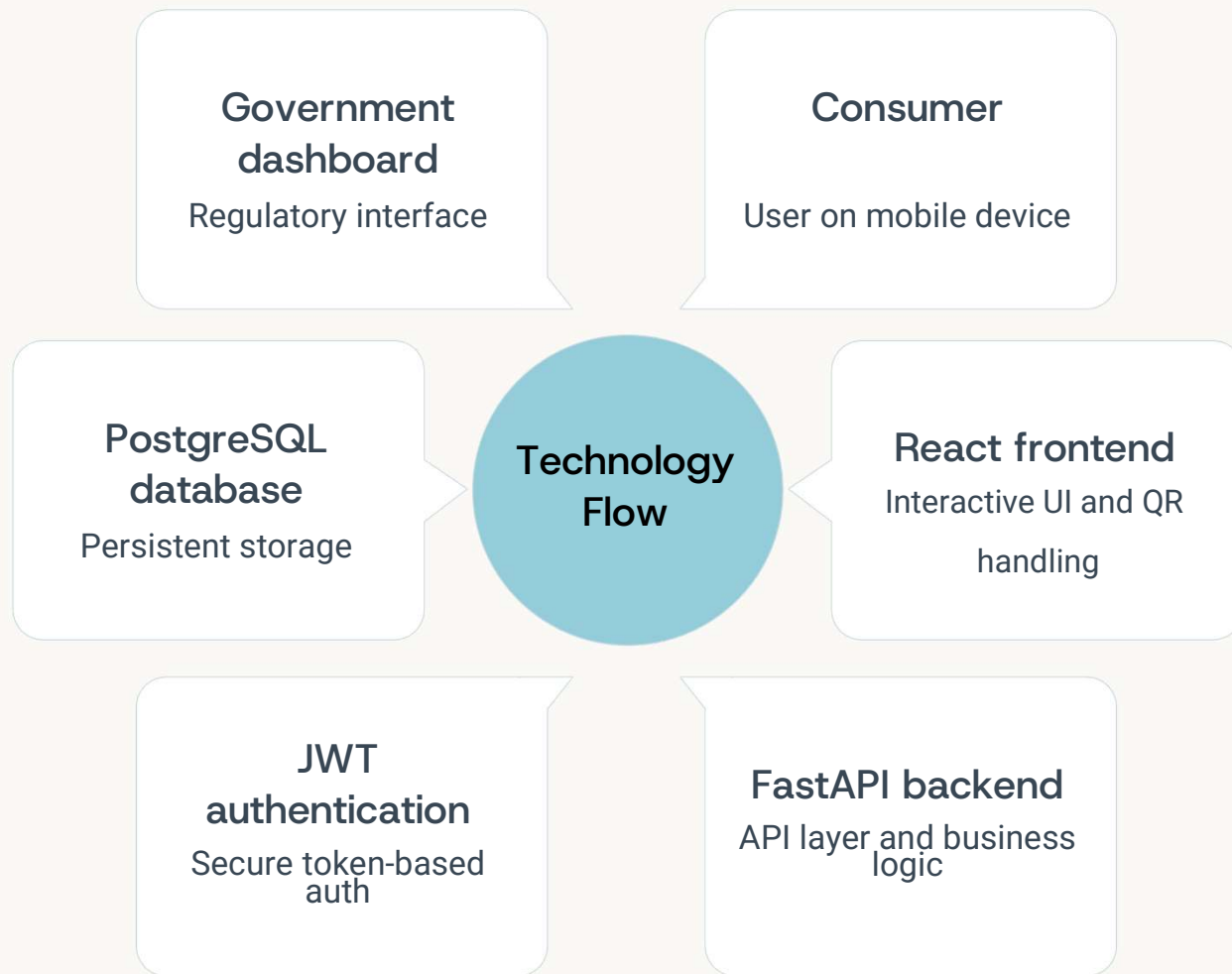
Transparency & Reporting

Digital audit trails and government reports improve oversight.



Stack: React frontend, FastAPI backend, JWT for security, PostgreSQL storage, QR code generation, and PDF reporting for audits.

System Architecture



Core Modules



Consumer Module

Aadhaar registration, QR, purchase history, limits, notifications, PDF/health reports, teetotaler mode.



Shop Operator Module

Login, QR scan, verify eligibility, sell products, sales history.



Government Admin Module

Dashboards, analytics, reports, policy enforcement tools.



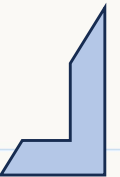
Doctor & Caretaker Modules

Medical restrictions, family monitoring, support workflows.

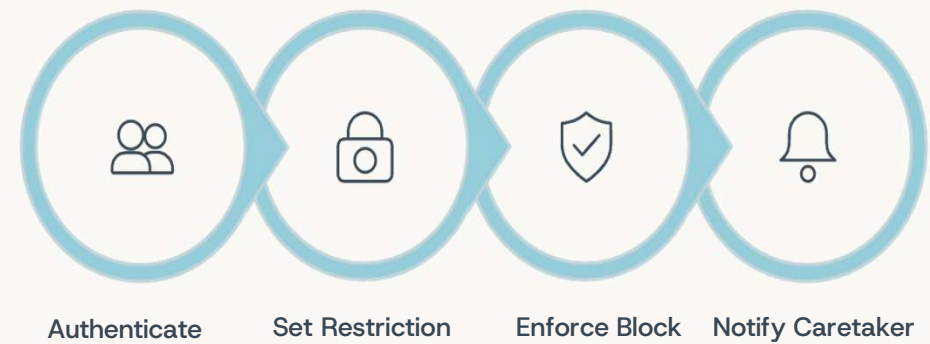
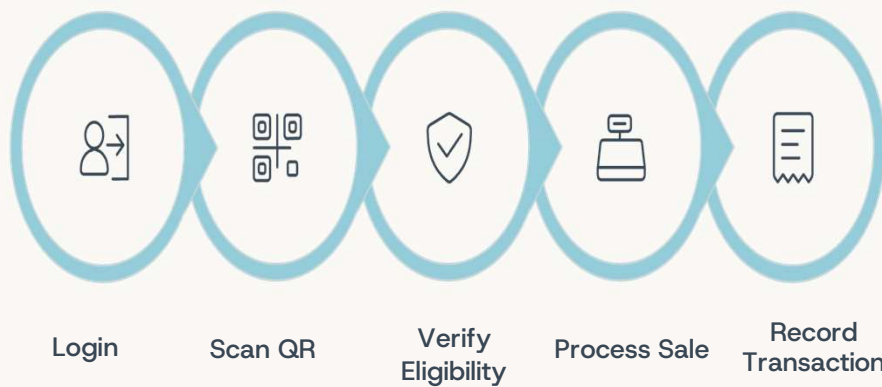


Smart TASMAL Consumer Management System – Module List

1. **Public Portal** – Landing page, registration, role selection.
2. **Consumer Module** – Registration, dashboard, QR code, purchase limits, history, reports.
3. **Shop Operator Module** – Consumer verification, QR scanning, sales management.
4. **Government Admin Module** – Consumer, shop, doctor, reports, analytics, system management.
5. **Doctor Module** – Issue restrictions, monitor patients, analytics.
6. **Caretaker Module** – Monitor linked consumer, receive alerts, manage consent.
7. **Authentication & Security Module** – Login, JWT authentication, role-based access.
8. **QR Management Module** – QR generation, validation, scanning.
9. **Notification Module** – SMS, email, and in-app alerts.
10. **Reports & Analytics Module** – Sales, revenue, addiction trends, PDF reports.
11. **Audit & Compliance Module** – Activity logs and transaction auditing.
12. **System Administration Module** – User management, settings, backup, monitoring.



Shop & Doctor Workflows



Both workflows integrate with the central backend to enforce limits and update audit logs in real time.

Key Smart Features



Aadhaar Verification



QR Code Purchase



Daily Purchase Limit



Weekly Limit



Monthly Limit



Doctor Restriction



Caretaker Monitoring



Notifications

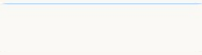


PDF Reports

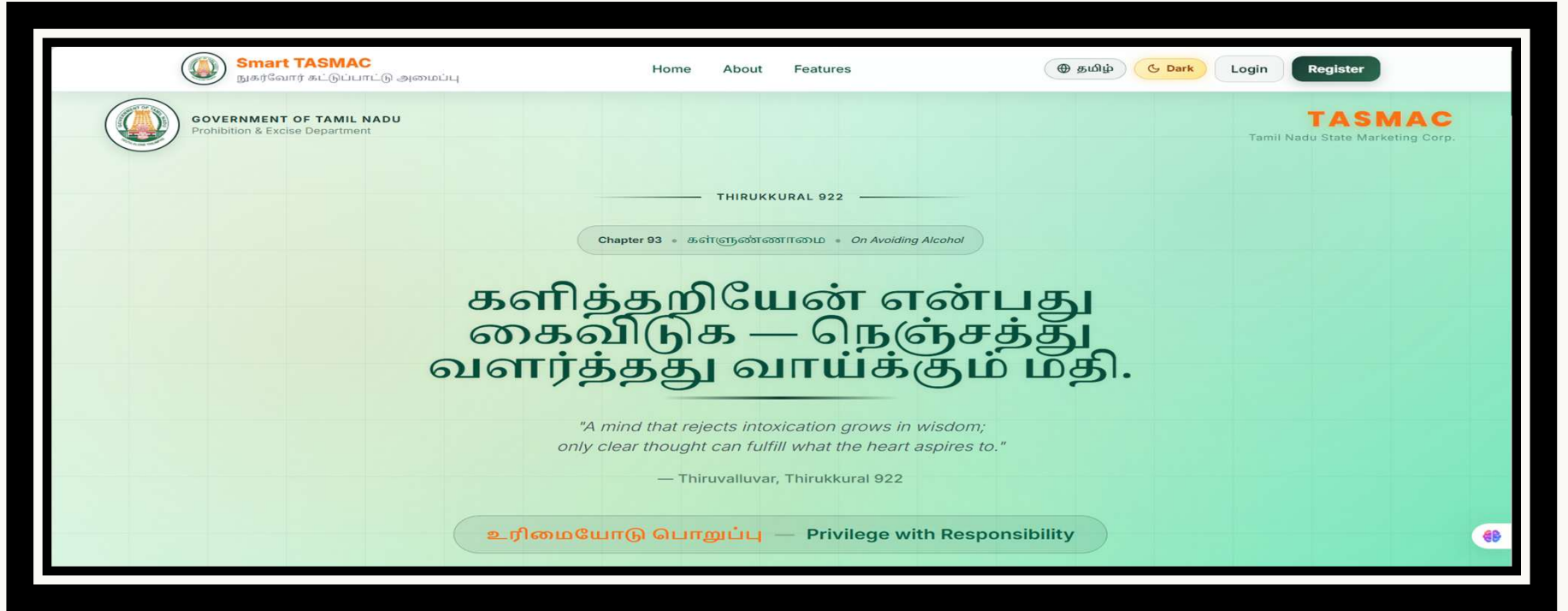


Audit Trail

Smart TASMAC offers a comprehensive suite of features to ensure responsible alcohol regulation, from identity verification to detailed analytics and medical oversight.



USER INTERFACE 1 : Landing portal for consumer



Public entry page for Smart TASMAC, offering system information, registration, and secure role-based login access.

USER INTERFACE 2 : Registration Portal for Consumer

How would you like to register?

Choose a method. You can switch between them at any time before submitting.



Manual Entry
Type your details directly. Works without a camera or scanner.
[Select >](#)



Upload ID Card (OCR)
Upload your Aadhaar card image — details are extracted automatically.
[Select >](#)

Already have an account? [Sign In](#)

1 YOUR DETAILS 2 REVIEW 3 SET PASSWORD 4 CONFIRM

Enter Your Details
Fill in your details manually. All fields marked * are required.

FULL NAME * As on Aadhaar card	DATE OF BIRTH * dd-mm-yyyy
GENDER * Male	AADHAAR NUMBER * 12-digit Aadhaar number
EMAIL ADDRESS * you@example.com	MOBILE NUMBER * 10-digit mobile number
DISTRICT * Select district...	
ADDRESS (OPTIONAL) Door no, street, city...	


Consumer Sign In
Smart TASMAC Consumer Management System

MOBILE OR AADHAAR NUMBER

Accepted digits **Saved info** X

PASSV

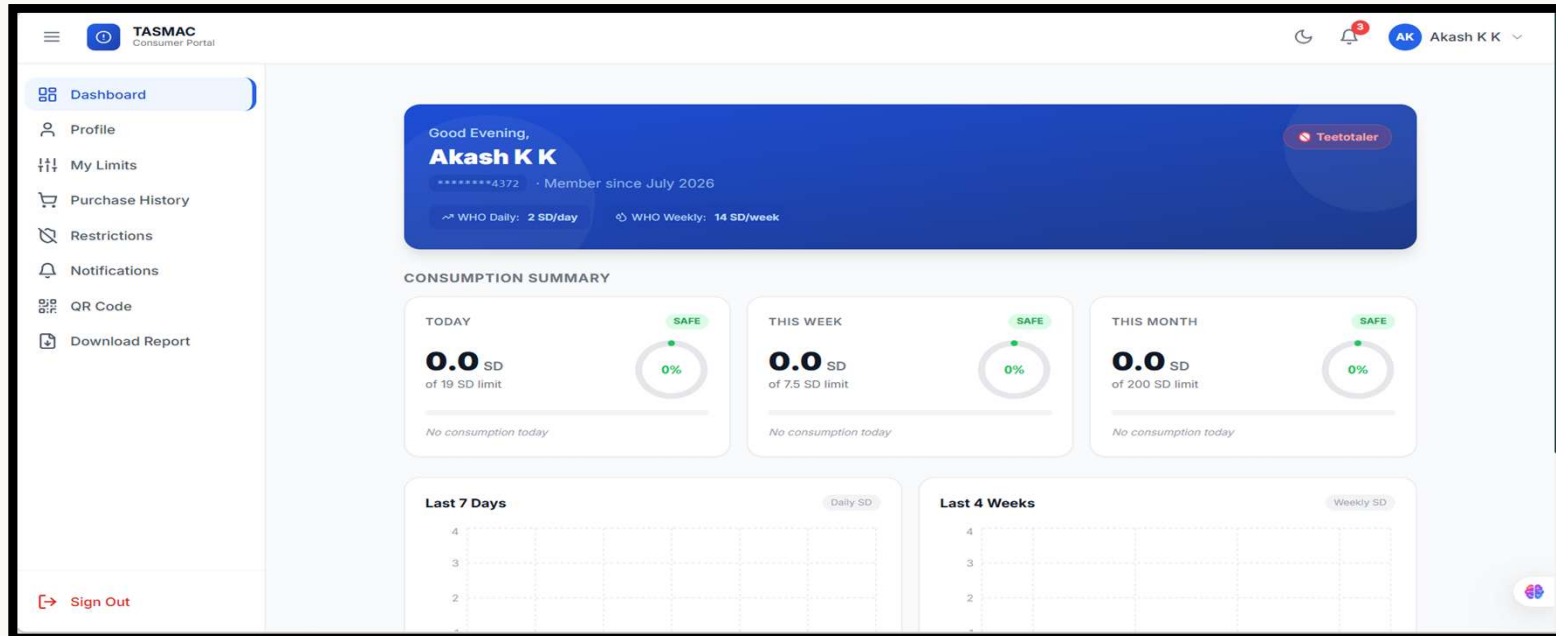
[-> Sign In](#)

New user? [Register here](#)

DEMO CREDENTIALS
Mobile: 9876543210
Password: Demo@1234pass

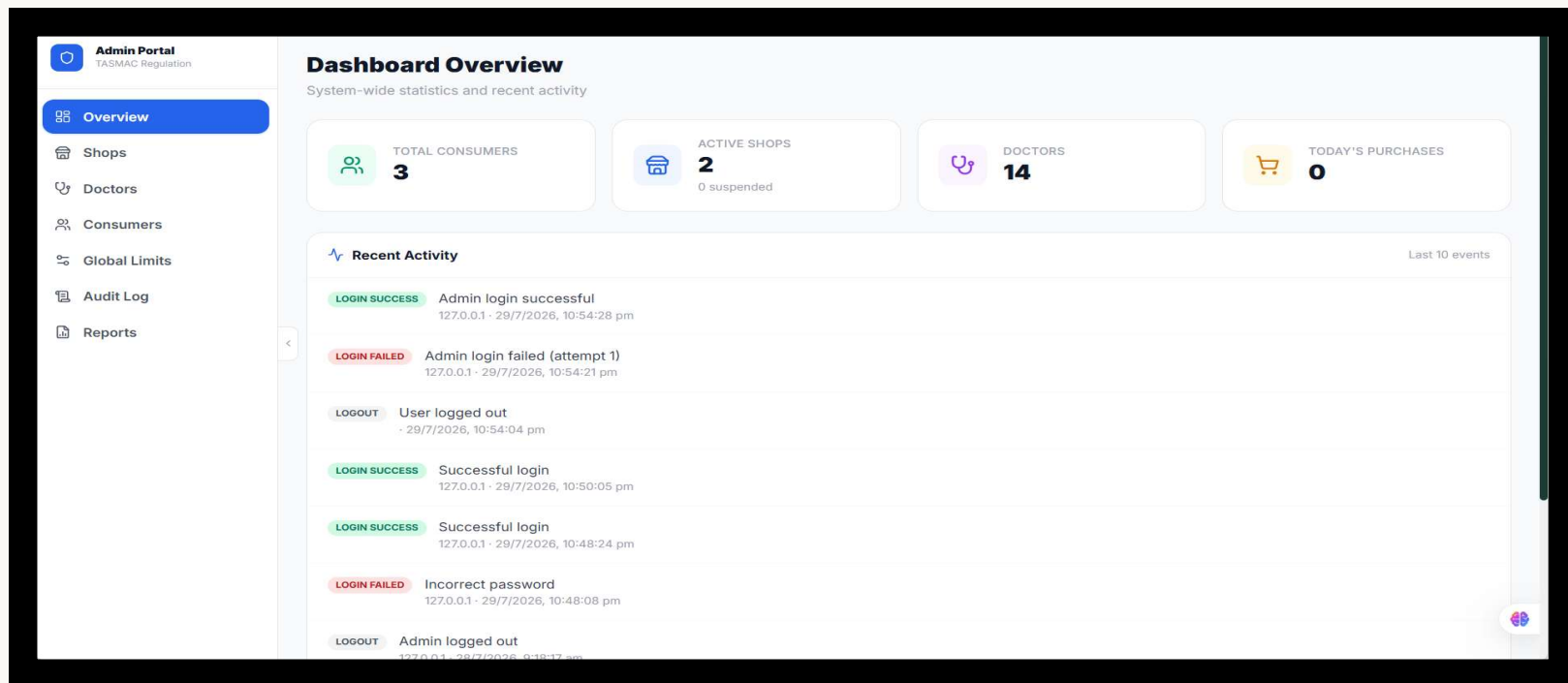
[Admin Portal](#) * [Shop Portal](#)

USER INTERFACE 3: Consumer Portal



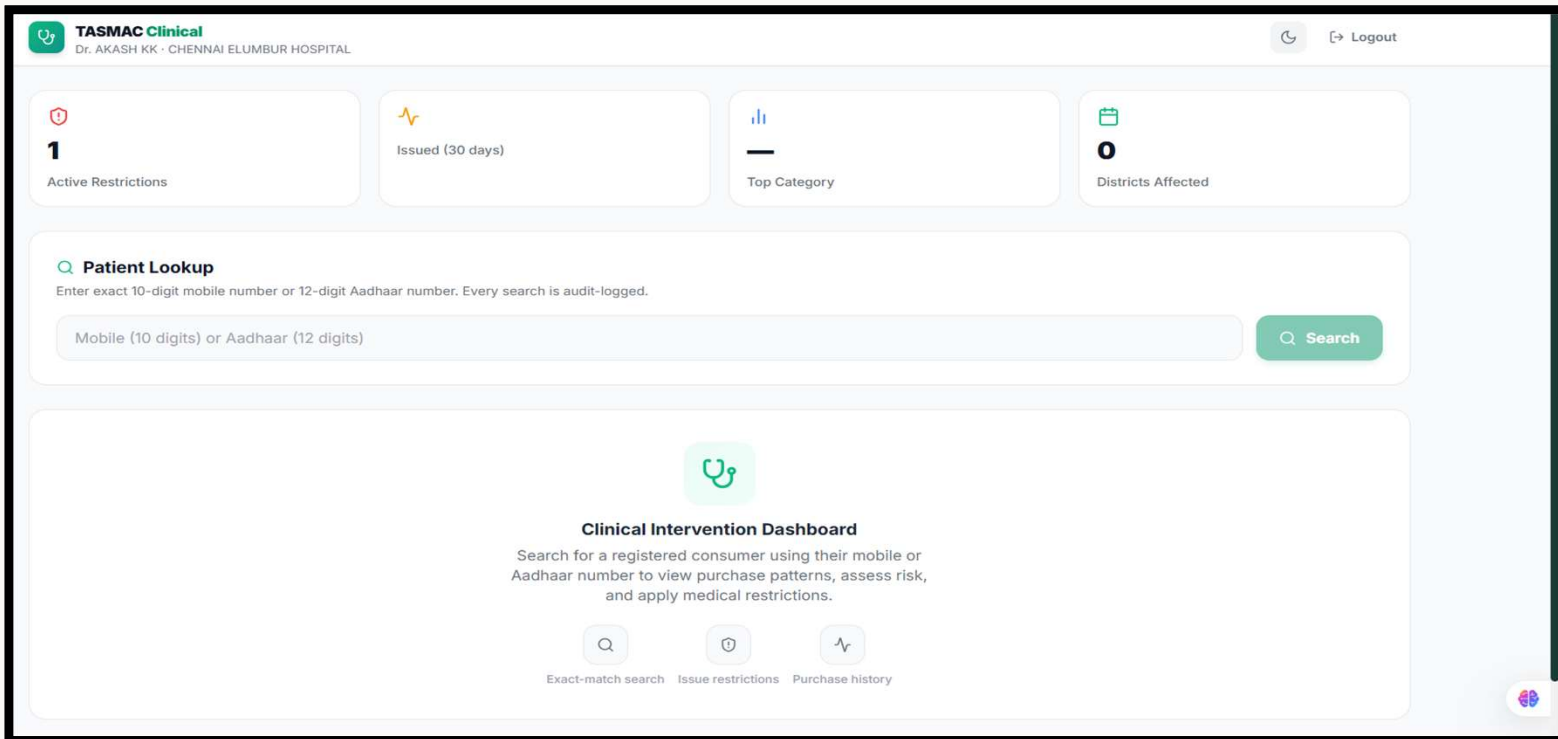
A personalized dashboard for consumers to track purchases, manage limits, access QR identity, and receive alerts.

USER INTERFACE 4: Admin Portal



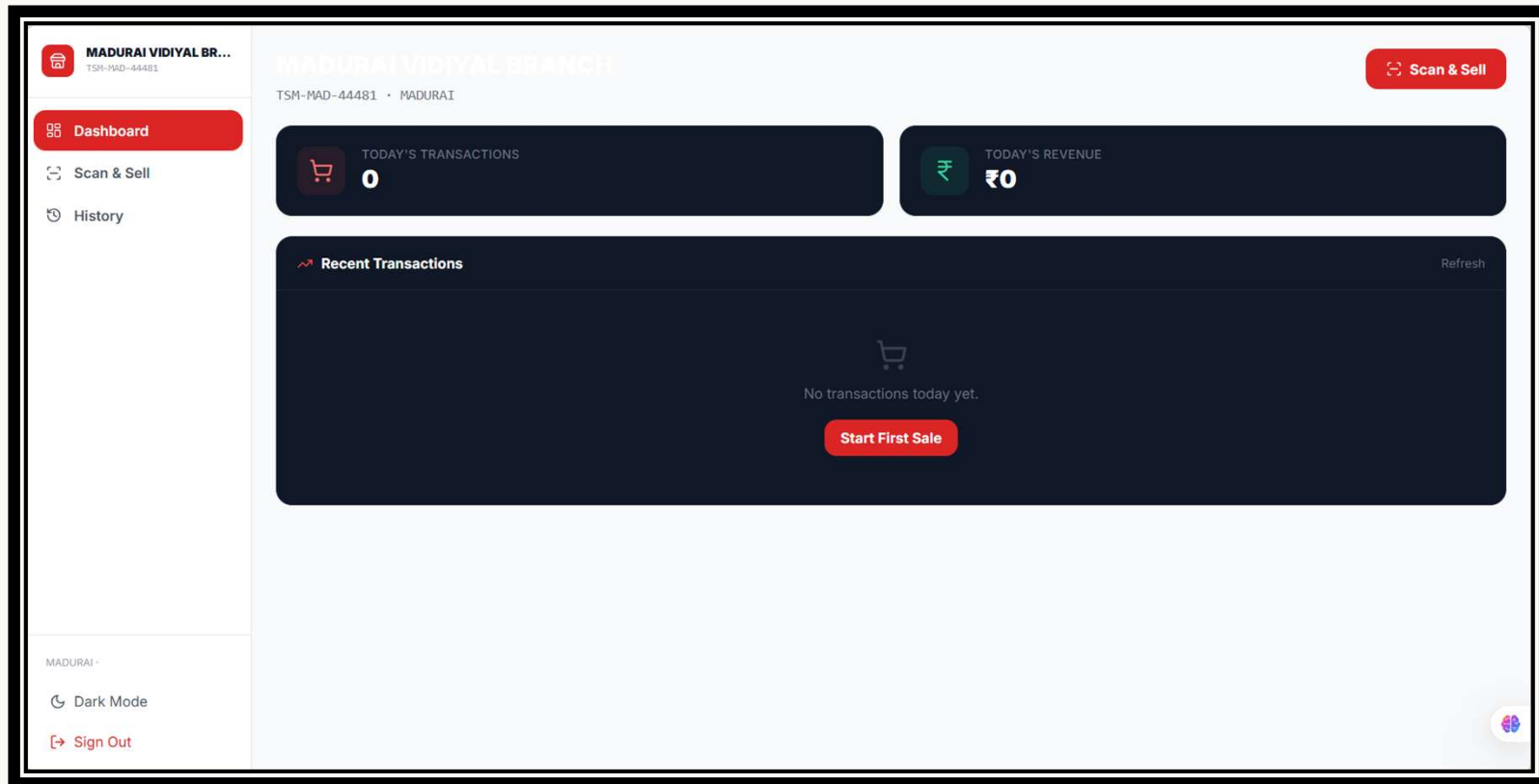
A centralized dashboard for managing users, shops, policies, reports, and monitoring the Smart TASMAC system.

USER INTERFACE 5: Doctor Portal



A secure portal for doctors to monitor consumer health, review purchase history, and apply medical restrictions

USER INTERFACE 6: Shop Portal



A secure portal for shop operators to verify consumers, process QR-based sales, and manage transaction records



Conclusion

Smart TASMAC is a secure, AI-enabled digital platform that promotes responsible alcohol consumption through identity verification, purchase regulation, healthcare integration, and real-time government monitoring.

Technology for Responsible Consumption • Transparency •
Public Health • Good Governance